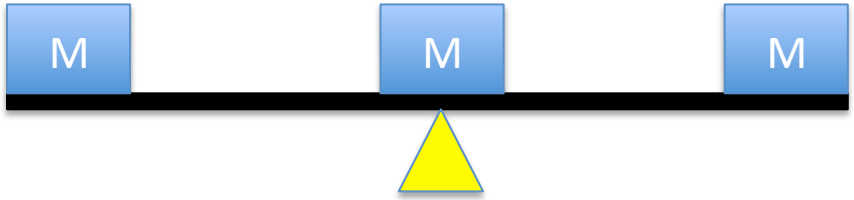
Levers

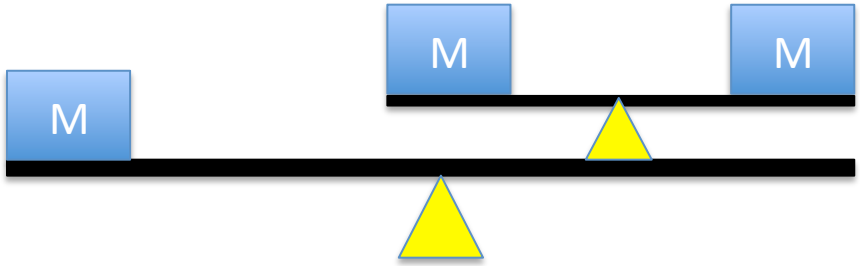
https://www.youtube.com/watch?v=YlYEi0PgG1g

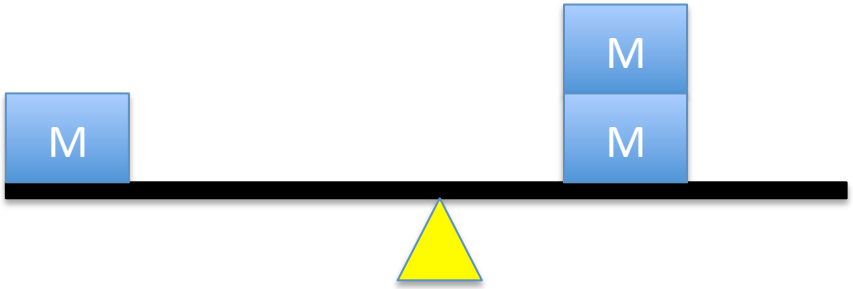
Levers

A lever is a rigid rod that rotates around one point to move a load by applying a force to a third point.

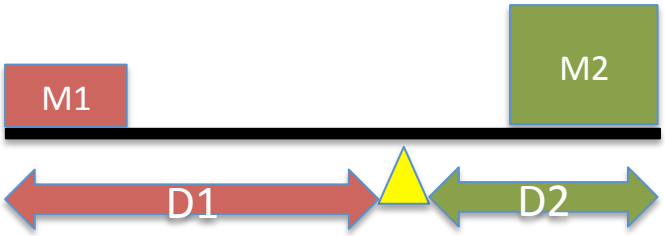
Archimedes worked out how things balance







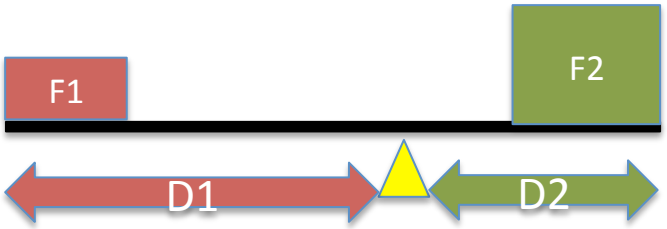
Balance occurs when moments are equal and opposite.



$$M_1D_1 = M_2D_2$$

Law of the Lever

$$F_1D_1 = F_2D_2$$



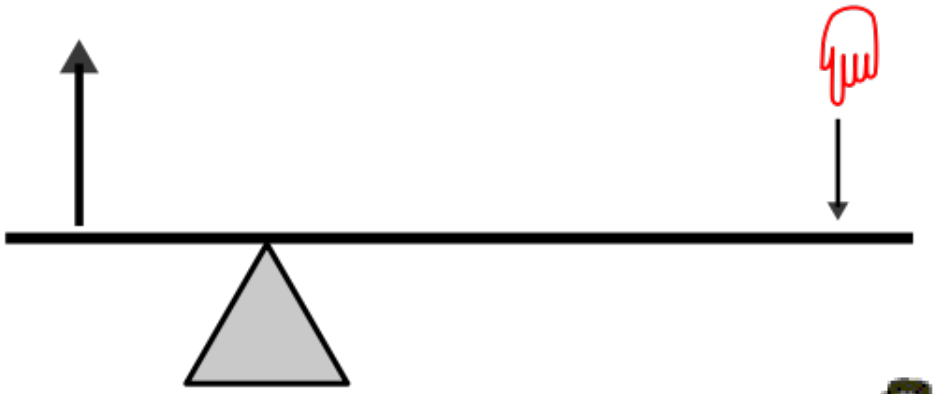
The masses are applying a downward force on the lever with their weight – which is a force. So the simple idea of balancing weights on a pivot reveals how forces act on a lever.

Three Classes of Levers

Classes of Levers

"First Class Lever"

- A first-class lever is a lever in which the fulcrum is located between the input effort and the output load.
- In operation, a force is applied (by pulling or pushing) to a section of the bar, which causes the lever to swing about the fulcrum, overcoming the resistance force on the opposite side.

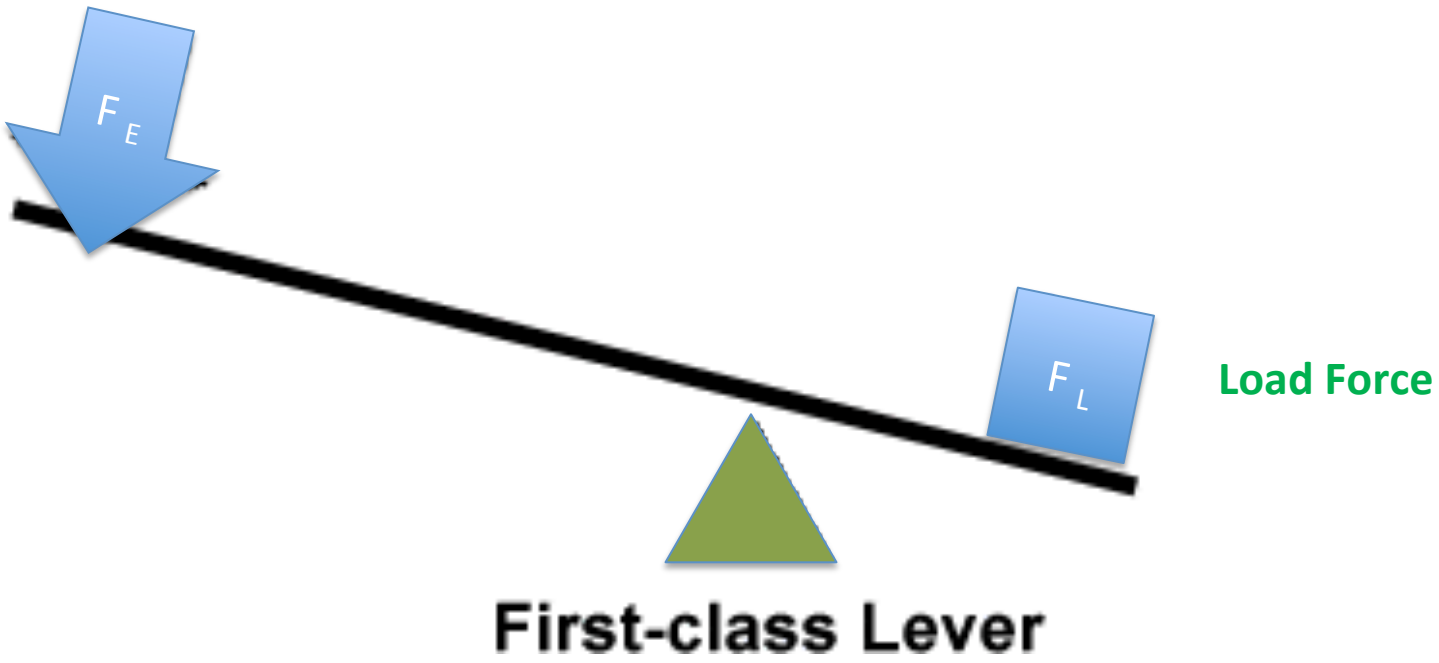


Examples:



- Seesaw
- Scissors (double lever)

First Class Lever



Fulcrum is between F_E (effort force) and F_L (load force)

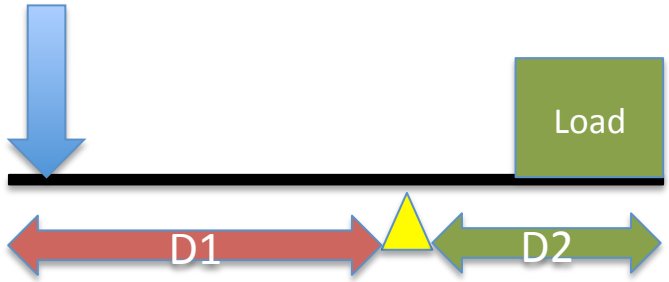
When the **effort moves farther than load**, the Mechanical advantage >1

When the **effort moves less than the load**, the Mechanical Advantage < 1

Lets apply the Law of the Lever

$$F_1D_1 = F_2D_2$$

Effort Force 10N



D1 = 50cm D2 = 10cm

What force of a load could be lifted?

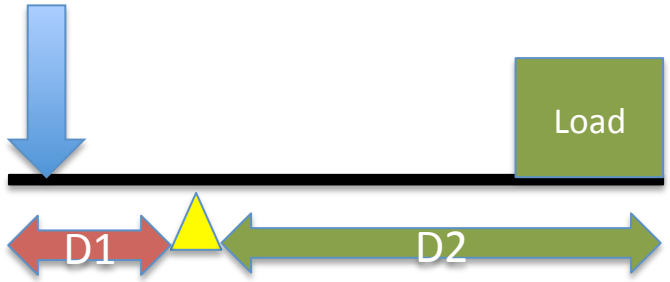
Mechanical Advantage = F_L/F_E

What is the Mechanical Advantage of this lever?

What happens if the effort is closer to the pivot than the load?

$$F_1D_1 = F_2D_2$$

Effort Force 10N



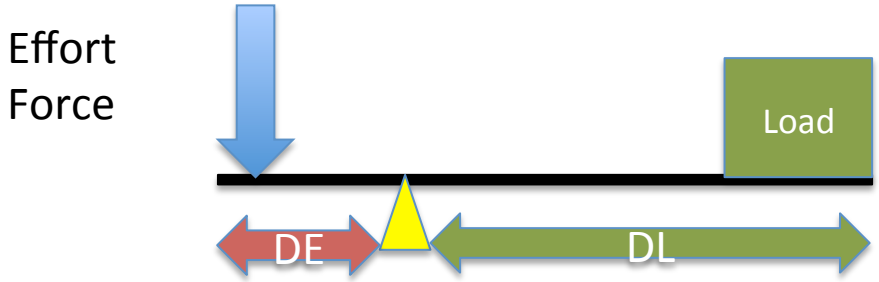
D1 = 10cm D2 = 50cm

Mechanical Advantage = F_L/F_E

What is the Mechanical Advantage of this lever?

What force of a load could be lifted?

The mechanical advantage of a lever is the ratio of the length of the lever on the applied effort side of the fulcrum to the length of the lever on the load force side of the fulcrum.



$MA = D_E/D_L$

Examples of first class levers

Common examples of first-class levers include ♪

- crowbars, ♪
- scissors, ♪
- pliers, ♪
- tin snips ♪
- and seesaws. ♪

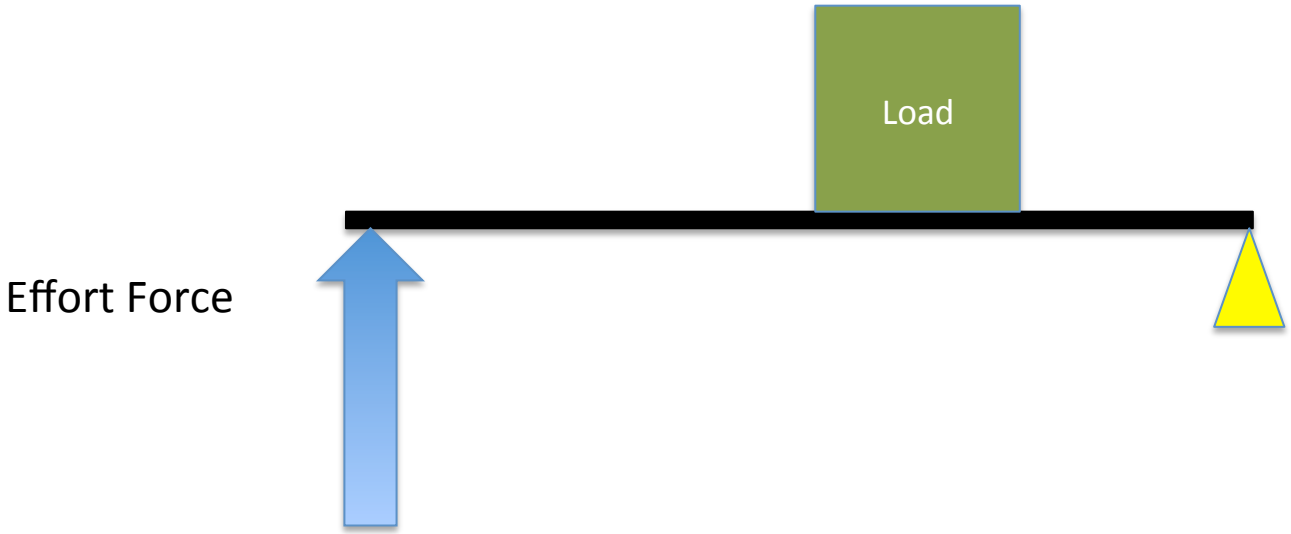
♪







Second Class Lever



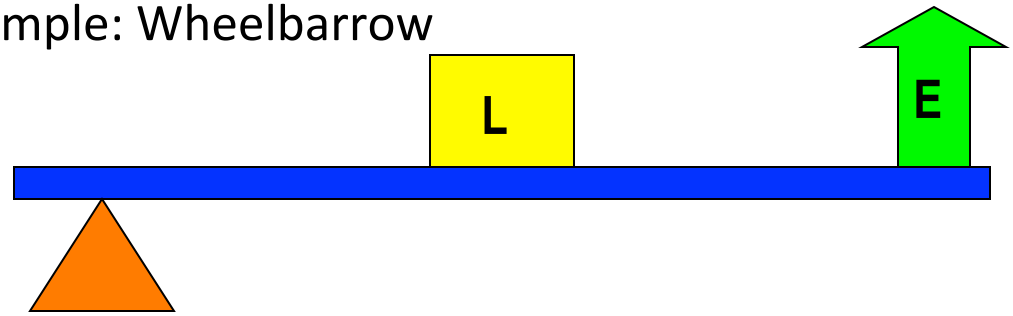
Load is between fulcrum and Effort
Effort moves farther than Load.

Multiplies Effort Force, but does not change its direction

The mechanical advantage of a 2nd class lever is always greater than 1

- Second class lever
  - Load is located between the effort force and the fulcrum.
  - Always multiplies a force

  - Example: Wheelbarrow



Always multiplies a force.

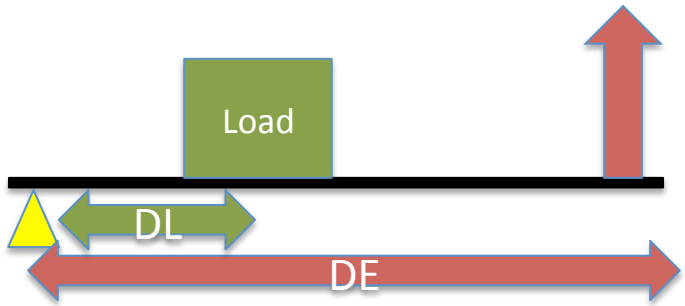
Examples of second-class levers

- Examples of second-class levers include:♪
- nut crackers, ♪
- wheel barrows, ♪
- doors, ♪
- and bottle openers.





Evaluate the Lever



Effort
Force
10N

DE = 80cm

DL = 20cm

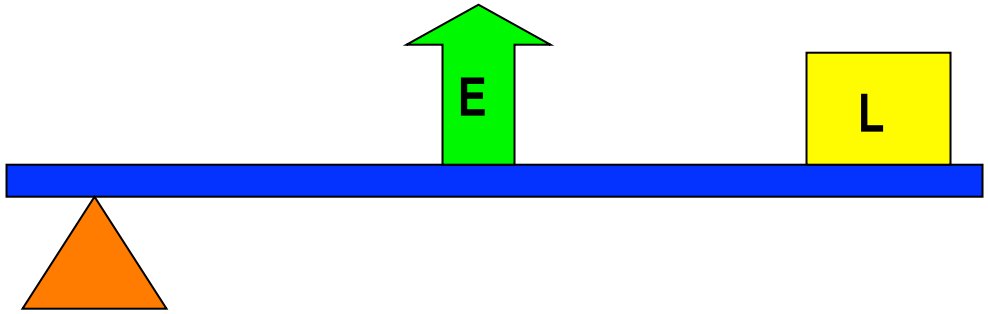
Mechanical Advantage = F_L/F_E

Mechanical Advantage = $D_{E/}D_L$

What force of a load could be lifted?

What is the Mechanical Advantage of this lever?

Third Class Lever



Effort is between fulcrum and Load.

Does not multiply force

Load moves farther than Effort.

Multiplies the distance the effort force travels and the speed at which it moves.

The mechanical advantage of a 3rd class lever is always less than 1.

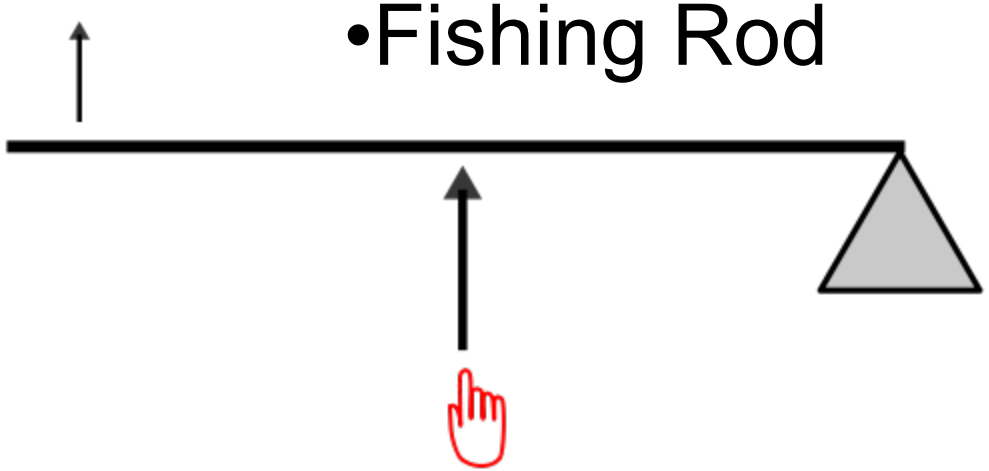
Classes of Levers

"Third Class Lever"

- For this class of levers, the input effort is higher than the output load, which is different from second-class levers and some first-class levers.
- However, the distance moved by the load is greater than the distance moved by the effort.
- In third class levers, effort is applied between the output load on one end and the fulcrum on the opposite end.

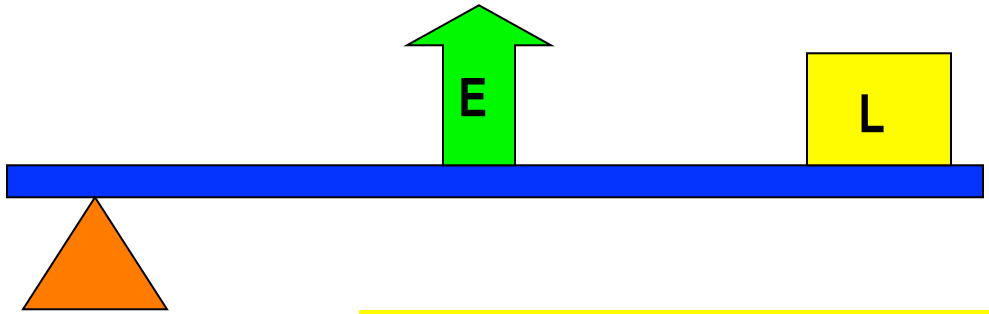
Examples:

- Hockey Stick
- Tweezers
- Fishing Rod



- Third class lever
  - Effort force located between the load and the fulcrum.
  - Effort arm is always shorter than load arm
  - MA is always less than one

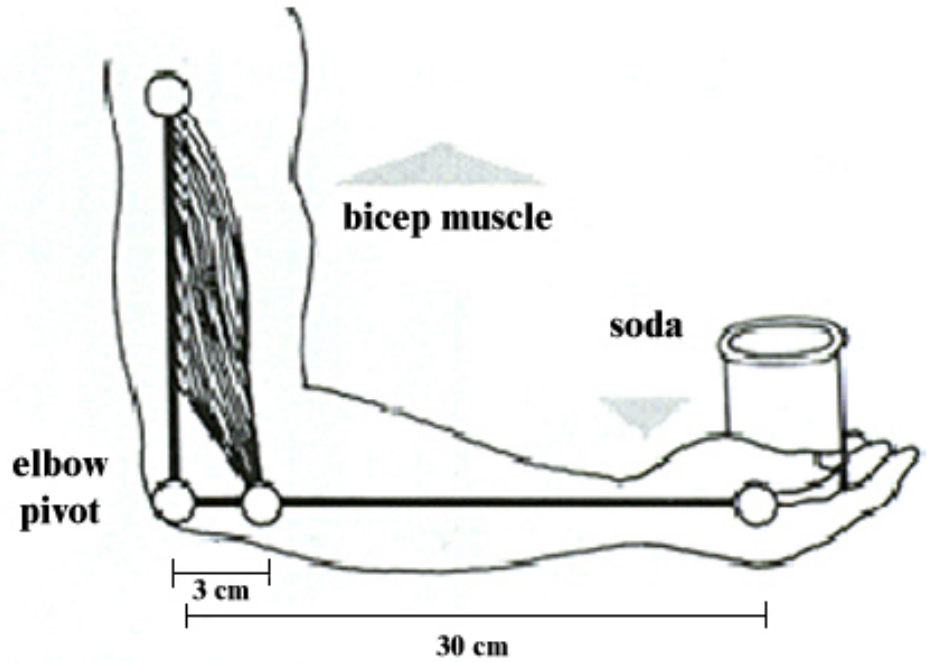
  - Example: Broom



There is an increase distance moved and speed at the other end. Other examples are baseball bat or hockey stick.

Examples of Third Class Levers♪

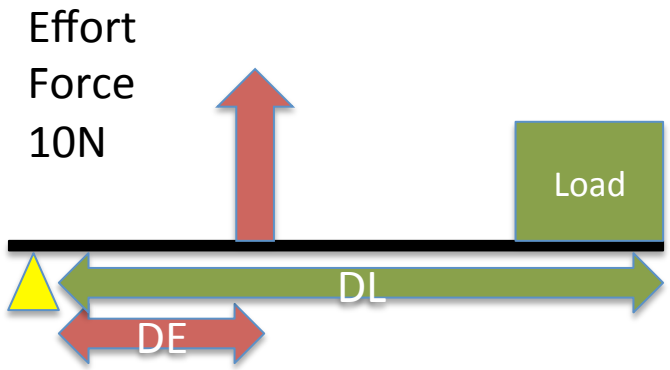
- Examples of third-class levers include:♪
  - tweezers, ♪
  - arm hammers, ♪
  - and shovels.♪





Third class lever in human body.

Evaluate the Lever



DE = 20cm

DL = 80cm

Mechanical Advantage = F_L/F_E

Mechanical Advantage = $D_{E/}D_L$

What force of a load could be lifted?

What is the Mechanical Advantage of this lever?

Mechanical Advantage

- Mechanical Advantage is the ratio between the load and effort.

- Mechanical Advantage deals only with forces.

- Mechanical Advantage **> 1** means that the output force will be greater than the input force.
  - (But the input distance will need to be greater than the output distance.)

Mechanical Advantage

- First and Second class levers have a positive mechanical advantage.
- Third class levers have a mechanical *disadvantage*, meaning you use more force that the force of the load you lift.

$$\text{MECHANICAL ADVANTAGE} = \frac{\text{EFFORT ARM}}{\text{RESISTANCE ARM}}$$

Movement Ratio

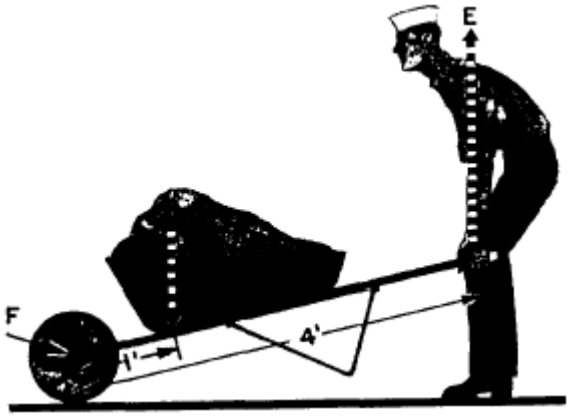
- Movement Ratio deals with the distance gained or lost due to a mechanical advantage.

- Movement Ratio >1 means that the input distance (or effort) to move a load will be greater than the output distance of the load.

Mechanical Advantage: Example

Mechanical Advantage =

$$\frac{\textit{effort arm}}{\textit{resistance arm}}$$



Crazy Joe is moving bricks to build his cabin.

With the use of his simple machine, a lever, he moves them easily.

The “effort arm” of his wheel barrow is 4ft., while the resistance arm of his wheelbarrow is 1 ft.